Alternating Direction Unfolding With a Cross Spectral Attention Prior for Dual-Camera Compressive Hyperspectral Imaging

Yubo Dong, *Member, IEEE*, Dahua Gao, *Member, IEEE*, Danhua Liu, *Member, IEEE*, Yanli Liu, and Guangming Shi, *Fellow, IEEE*

***Abstract*—Coded Aperture Snapshot Spectral Imaging (CASSI) multiplexes 3D Hyperspectral Images (HSIs) into a 2D sensor to capture dynamic spectral scenes, which, however, sacrifices the spatial information. Dual-Camera Compressive Hyperspectral Imaging (DCCHI) enhances CASSI by incorporating a Panchromatic (PAN) camera to compensate for the loss of spatial information in CASSI. However, the dual-camera structure of DCCHI disrupts the diagonal property of the product of the sensing matrix and its transpose, making it difficult to efficiently and accurately solve the data subproblem in closed-form and thereby hindering the application of model-based methods and Deep Unfolding Networks (DUNs) that rely on such a closed-form solution. To address this issue, we propose an Alternating Direction DUN, named ADRNN, which decouples the imaging model of DCCHI into a CASSI subproblem and a PAN subproblem. The ADRNN alternately solves data terms analytically and a joint prior term in these subproblems. Additionally, we propose a Cross Spectral Transformer (XST) to exploit the joint prior. The XST utilizes cross spectral attention to exploit the correlation between the compressed HSI and the PAN image, and incorporates Grouped-Query Attention (GQA) to alleviate the burden of parameters and computational cost brought by impartially treating the compressed HSI and the PAN image. Furthermore, we built a real DCCHI system and captured large-scale indoor and outdoor scenes for future academic research. Extensive experiments on both simulation and real datasets demonstrate that the proposed method achieves state-of-the-art (SOTA) performance. The code and datasets have been open-sourced at: https://github.com/ShawnDong98/ADRNN-XST**

***Index Terms*—Compressive spectral imaging, deep unfolding networks, large-scale real data.**

I. Introduction

HYPERSPECTRAL imaging, which captures spectral data across various bands, is instrumental in many fields, including biomedical research [32], remote sensing [60], [61], and industrial inspection [24]. Traditional hyperspectral imaging systems obtain a 3D Hyperspectral Image (HSI) by scanning the scene through multiple exposures, making them unsuitable for dynamic scenes. To overcome this limitation, Coded Aperture Snapshot Spectral Imaging (CASSI) [45] compresses the 3D HSI data into a 2D spatial sensor, enabling the capture of dynamic scenes but at the expense of quality spatial information.

The CASSI system trades off degradation of 2D spatial information to preserve potential 3D spectral information. There are two-fold spatial degradation in CASSI: 1) the coded aperture disrupts the spatial information, 2) the prism and detector disperse and multiplex light across different wavelengths, which degrades the edge information of the scene.

Dual-Camera Compressive Hyperspectral Imaging (DCCHI) [47], [48] augments the conventional CASSI with the addition of a Panchromatic (PAN) camera to mitigate the spatial information loss associated with the CASSI system. The choice to include an extra PAN branch has three merits. The PAN branch eliminates the need for complex Image Signal Processing (ISP) pipelines, making it simpler than the RGB branch [18], [20]. Compared to the compressive multi-spectral branch [14], the PAN branch also preserves superior spatial texture details by avoiding degradation caused by encoding, dispersion, and multiplexing. Lastly, the PAN branch employs the same detector as the CASSI branch, resulting in similar spectral response curves, which brings potential benefits to reconstruction processes. Similar choice also has been adopted by previous work [62]. DCCHI thereby gains the ability to recover 3D HSI by exploiting the synergy between the compressed HSI and the PAN image, revealing considerable promise for real-time hyperspectral imaging applications. Nevertheless, the primary challenge lies in accurately restoring the high-fidelity, intrinsic 3D HSI from the compressed HSI and the PAN image.

Recently, various reconstruction methods have been developed for the DCCHI system, including traditional model-based methods [4], [11], [20], [27], [49], [55], end-to-end neural networks [6], [31], [34], and Deep Unfolding Networks (DUNs) [7], [16], [17], [22], [23], [53], [59]. Among them, DUNs have gained significant attention as they integrate the advantages of model-based methods and end-to-end neural networks. DUNs are heuristically designed by unfolding model-based methods into a stack of Deep Neural Networks (DNNs), which alternate between solving a data subproblem and a prior subproblem.

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Yubo Dong, Dahua Gao, Danhua Liu, and Guangming Shi are with the School of Artificial Intelligence, Xidian University, Xi'an 710126, China (e-mail: ybdong@xidian.edu.cn; dhgao@xidian.edu.cn; dhliu@xidian.edu.cn; gmshi@xidian.edu.cn).

Yanli Liu is with Beijing Institute of Space Mechanics and Electricity, Beijing 100086, China (e-mail: ylliu214@126.com).

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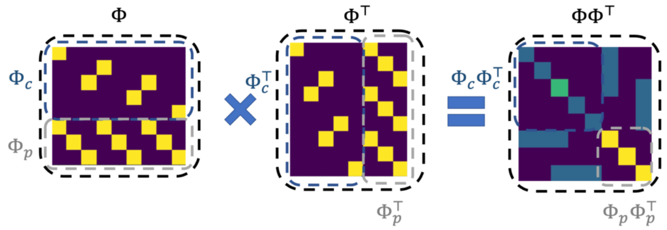

Fig. 1. Diagram of the diagonal property of the product of the sensing matrix and its transpose. The blue box illustrates the CASSI system, the gray box illustrates the PAN system, and the black box illustrates the DCCHI system.

To efficiently and accurately solve the data subproblem in closed-form, it is essential that the product of the sensing matrix and its transpose has a diagonal structure.

In DCCHI, as shown in Fig. 1, the dual-camera structure disrupts the diagonal property of the product of the sensing matrix Φ and its transpose $\Phi^{\top}$, making it difficult to solve the data subproblem in closed-form, hindering the application of model-based methods and DUNs that rely on a closed-form solution, such as Alternating Direction Method of Multipliers (ADMM) [5], [28], Generalized Alternating Projection (GAP) [26], [30], [55], Half-Quadratic Splitting (HQS) [7], [16], [19], [58] based methods. Some previous methods [47], [50] use gradient descent to circumvent this issue. However, compared to the closed-form solution, gradient descent not only converges more slowly but also yields a less accurate solution. One possible way is to employ the CASSI imaging model to derive the closed-form solution of the data subproblem, using the PAN image only as guiding information in solving the prior subproblem. However, this approach presents two issues: 1) the imaging model does not match the imaging process, and 2) it fails to leverage the information of the PAN image in solving the data subproblem. To address these issues, we propose an Alternating Direction DUN, named ADRNN, which decouples the imaging model of DCCHI into a CASSI subproblem and a PAN subproblem. Therefore, ADRNN could leverage diagonal matrix $\Phi_c\Phi_c^{\top}$ in CASSI subproblem, and $\Phi_p\Phi_p^{\top}$ in PAN subproblem, to alternately solve data subproblems analytically.

To solve the joint prior subproblem, it is crucial to effectively exploit the correlation between the compressed HSI and the PAN image. Most previous methods [11], [20], [49] have employed the PAN image as prior information to guide the reconstruction of HSI. Nevertheless, these methods have primarily focused on utilizing the PAN image to enhance the spatial information of the compressed HSI, yet they have neglected to utilize the compressed HSI to reciprocally enrich the spectral information of the PAN image. To address these issues, we propose a cross spectral attention mechanism to exploit the correlation between the HSI and the PAN image, treating the compressed HSI and the PAN image impartially. Additionally, we introduce the Grouped-Query Attention (GQA) [2] to alleviate the burden of parameters and computational cost brought by impartially treating the compressed HSI and the PAN image.

In order to make progress on the reconstruction problem, we reproduced a real DCCHI system and collected a variety of compressive HSIs and PAN images from indoor and outdoor scenes, serving as a benchmark for future DCCHI research. The built DCCHI system disperses horizontally a pixel at 463 nm, 466 nm, and so on, across 40 different wavelengths in the range of 463 nm to 684 nm. Consequently, for each measurement pair, a spatial-spectral 3D cube with 40 channels can be reconstructed.

Our contribution can be summarized as follows:

- We propose an alternating direction unfolding framework, named ADRNN, which addresses the challenge of solving the data subproblem in closed-form in the DCCHI system by decoupling the imaging model of DCCHI into a CASSI subproblem and a PAN subproblem.
- A Cross Spectral Transformer (XST) is proposed to exploit the joint prior of ADRNN, utilizing cross spectral attention to exploit the correlation between the compressed HSI and the PAN image, and incorporating GQA to alleviate the burden of parameters and computational cost brought by impartially treating the compressed HSI and the PAN image.
- We have built a real DCCHI system and captured a dataset from indoor and outdoor scenes. The dataset will be released to the academic community for future research.
- The proposed ADRNN-XST achieves state-of-the-art (SOTA) performance on both simulation and real datasets.

II. Related Works

A. Deep Unfolding Networks

Traditional model-based methods formulate an optimization problem incorporating prior regularization of the HSI, such as Total Variation (TV) minimization [4], [11], [55] and the low-rank approach [20], [27]. These methods are usually decoupled into a data fidelity term and a prior term, resulting in an iterative process that alternates between solving a data subproblem and a prior subproblem. While these methods offer strong theoretical guarantees and high interpretability, they often suffer from the limited representativeness of hand-crafted priors, leading to poor reconstruction quality.

To address the above issue, plug-and-play (PnP) methods [9], [10], [29], [33], [52], [56] adopt a pre-trained denoiser to implicitly express the prior subproblem in model-based methods as a denoising problem. This approach allows the integration of advanced deep learning capabilities into traditional model-based frameworks, providing a balance between interpretability and performance.

In contrast to PnP methods, DUNs [7], [16], [17], [22], [23], [46], [50] use a unique trainable denoiser at each stage rather than a shared pre-trained denoiser across all stages and have achieved great success in HSI reconstruction.

All aforementioned methods require solving a data subproblem using gradient descent [4], [17], [46], [50] or a closed-form solution [7], [11], [16], [27], [55]. Compared to gradient descent, the closed-form solution is typically more accurate and converges faster.

To efficiently obtain the closed-form solution is highly dependent on the property that the product of the sensing matrix and its transpose is a diagonal matrix. However, the dual-camera structure of the DCCHI system disrupts this diagonal property, making it difficult to efficiently and accurately solve the data subproblem in closed-form, thereby hindering the application of methods that rely on such a closed-form solution.

B. Cross Attention Mechanism

The concept of cross attention was initially popularized by the Transformer architecture, particularly in the context of nat-

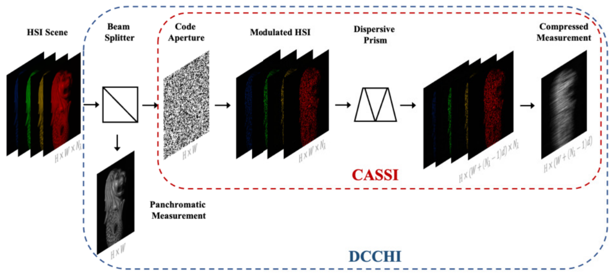

Fig. 2. Diagram of CASSI system and DCCHI system.

ural language processing (NLP) [44]. In standard Transformer decoders, cross attention is used to attend to all positions of the input sequence, effectively capturing dependencies across different parts of the sequence.

Later, cross attention mechanisms were widely explored in various multimodal transformer models, especially in the context of vision-and-language tasks. LXMERT [41] and UNITER [8] leveraged cross attention to align and integrate information from different modalities, significantly improving performance on multimodal tasks.

Recently, the cross attention mechanism has been adapted to image processing tasks to handle dependencies across different views or different parts of images. NAFSSR [13] has achieved competitive performance in stereo super-resolution by adding cross attention modules to fuse features between views, fully utilizing both intra-view information and cross-view information.

The compressed HSI and the PAN image can be treated as different modalities or different views of the same scene, making it suitable to utilize cross attention mechanism to exploit the correlation between the compressed HSI and the PAN image.

C. Grouped-Query Attention Mechanism

In recent years, large language models (LLMs) [1], [42], [43] have made significant progress. However, the computational efficiency and memory bandwidth overhead of LLMs have hindered their inference speed. Shazeer [40] pioneered the idea of Multi-Query Attention (MQA) to mitigate these issues by using multiple query heads while sharing a single key and value head across all queries. This approach showed significant benefits, particularly for models handling long inputs. Ainslie et al. [2] developed the GQA mechanism, which interpolates between Multi-Head Attention (MHA) [44] and MQA by grouping query heads and sharing single key and value heads within each group. This approach has been shown to offer a favorable trade-off between the quality of MHA and the speed of MQA.

GQA has not yet been fully explored in visual tasks. Therefore, we aim to utilize GQA to alleviate the burden of parameters and computational cost of XST brought by treating the compressed HSI and the PAN image impartially. Additionally, the GQA compresses the knowledge of MHA into limited parameters, thereby enhancing the representation capability of networks.

III. Mathematical Model of Observation

In the CASSI system [14], [45], as shown in the red box in Fig. 2, the incident light is initially projected into the coded aperture to perform a spatial modulation. Then, the modulated incident light is spectrally dispersed by the prism. Finally, the spectrally dispersed information is captured by a grayscale camera. Essentially, CASSI trades off degradation of 2D spatial information to preserve potential 3D spectral information.

Let's denote an HSI signal tensor $X \in \mathbb{R}^{H\times W\times N_\lambda}$, and a coded aperture tensor $M \in \mathbb{R}^{H\times W}$, where N_λ represents the number of wavelengths. The image that results from the modulation process at the n_λ^{th} wavelength can be depicted in the following manner:

$$X_c'(:,:,n_\lambda) = M \odot X(:,:,n_\lambda), \tag{1}$$

where $\odot$ represents the element-wise multiplication. The coded aperture disrupts the spatial information. Subsequently, the spatially modulated HSI X_c' is spectrally dispersed by the dispersive prism, which can be expressed as

$$X_c''(h, w+d_{n_\lambda}, n_\lambda) = X_c'(h,w,n_\lambda), \tag{2}$$

where d_{n_λ} denotes the dispersed distance of the n_λ-th wavelength, and $X_c'' \in \mathbb{R}^{H\times(W+d_{N_\lambda})\times N_\lambda}$. At last, the grayscale camera captures the dispersed HSI into a 2D measurement, which can be formulated as

$$Y_c = \sum_{n_\lambda=1}^{N_\lambda} X_c''(:,:,n_\lambda), \tag{3}$$

where $Y_c \in \mathbb{R}^{H\times(W+d_{N_\lambda})}$. The dispersion process expands the spatial dimensions of the signal, while the 2D imaging process compresses its spectral dimension. The prism and detector disperse and multiplex light across different wavelengths, which degrades the edge information of the scene.

In terms of matrix-vector form, Eq. (3) can be represented as

$$y_c = \Phi_c x, \tag{4}$$

where $x \in \mathbb{R}^{HWN_\lambda}$ is the original HSI. $y_c \in \mathbb{R}^{HW'}$ is the CASSI measurement (compressed HSI). $\Phi_c \in \mathbb{R}^{HW'\times HWN_\lambda}$ is the CASSI sensing matrix, typically treated as the shifted mask.

In the DCCHI system [47], as shown in the blue box in Fig. 2, the incident light is first divided into two directions by the beam splitter. The light in one direction is captured by CASSI, while the light in another direction is captured by the same kind of grayscale camera. The discretized form of PAN measurement can be represented as

$$Y_p = \sum_{n_\lambda=1}^{N_\lambda} X(:,:,n_\lambda), \tag{5}$$

Eq. (5) can also be rewritten in matrix-vector form

$$y_p = \Phi_p x, \tag{6}$$

where $y_p \in \mathbb{R}^{HW}$ is the PAN measurement (PAN image). $\Phi_p \in \mathbb{R}^{HW\times HWN_\lambda}$ is the PAN sensing matrix, denoting the projection matrix of the PAN camera.

The DCCHI system sensing process can be generally expressed as

$$(y_c, y_p) = (\Phi_c, \Phi_p)x + g, \tag{7}$$

where $g \in \mathbb{R}^{H(W'+W)}$ represents the measurement noise.

The aim is to restore the high-quality image x from the CASSI measurement y_c and the PAN measurement y_p, which is typically an ill-posed problem.

IV. Method

A. Algorithms Relying on a Closed-Form Solution

In this section, we will use the Half-Quadratic Splitting (HQS) algorithm as an example to illustrate the derivation process of the algorithms that rely on a closed-form solution and the issues encountered in applying these algorithms in the DCCHI system.

Most traditional model-based methods [4], [5], [19], [27] and DUNs [7], [16], [17], [22], [28], [30], [59] are based on the theory of maximum a posteriori (MAP). Specifically, the original HSI signal could be estimated by minimizing the following energy function as:

$$\hat{x} = \arg\min_x \frac{1}{2}\|y - \Phi x\|_2^2 + \lambda J(x), \tag{8}$$

where $\frac{1}{2}\|y-\Phi x\|_2^2$ represents the data fidelity term, $J(x)$ denotes the image prior term, and λ is a hyperparameter that balances their importance. By introducing an auxiliary variable z, Eq. (8) can be reformulated as:

$$\hat{x} = \arg\min_x \frac{1}{2}\|y - \Phi x\|_2^2 + \lambda J(z), \quad s.t. \quad z = x. \tag{9}$$

This is a constrained optimization problem. We adopt the HQS algorithm for speedy and efficient unfolding. Consequently, Eq. (9) can be solved by minimizing:

$$L_\mu(x, z) = \frac{1}{2}\|y - \Phi x\|_2^2 + \lambda J(z) + \frac{\mu}{2}\|z - x\|_2^2, \tag{10}$$

where μ serves as a penalty parameter that forces x and z to approach the same fixed point. Subsequently, Eq. (10) can be solved by decoupling x and z into the following two iterative subproblems as:

$$x^{k+1} = \arg\min_x \|y - \Phi x\|_2^2 + \mu\|x - z^k\|_2^2, \tag{11a}$$

$$z^{k+1} = \arg\min_z \frac{\mu}{2}\|z - x^{k+1}\|_2^2 + \lambda J(z), \tag{11b}$$

where $k = 0, 1, \ldots, K-1$ indexes the iteration. Note that the data fidelity term is associated with a quadratic regularized least-squares problem, and it has a closed-form solution:

$$x^{k+1} = (\Phi^\top \Phi + \mu I)^{-1}(\Phi^\top y + \mu z^k). \tag{12}$$

Since the matrix $\Phi^\top\Phi + \mu I$ is very large, directly computing its inverse becomes computationally prohibitive [15], [37], [38], [46]. However, thanks to the specific structures of the sensing matrices Φ_c and Φ_p in CASSI and PAN systems, respectively, these systems can leverage recent advances in efficient computation methods [27], [55]. In these systems, the product $\Phi\Phi^\top$ turns out to be a diagonal matrix:

$$\Phi\Phi_c^\top = \mathrm{diag}\{\phi^1, \ldots, \phi^i, \ldots, \phi^{HW'}\}, \tag{13}$$

Following the matrix inverse lemma, the matrix inversion in Eq. (12) can be written as follows:

$$(\Phi^\top\Phi + \mu)^{-1} = \mu^{-1}I - \mu^{-1}\Phi^\top(I + \Phi\mu^{-1}\Phi^\top)^{-1}\Phi\mu^{-1}. \tag{14}$$

By plugging $\Phi\Phi^\top$ into $(I + \Phi\mu^{-1}\Phi^\top)^{-1}$, we obtain:

$$\begin{aligned}(I + \Phi\mu^{-1}\Phi^\top)^{-1} &= \mathrm{diag}\left\{\frac{\mu}{\mu + \phi_c^1}, \ldots, \frac{\mu}{\mu + \phi^{HW'}}\right\} \\ &= \frac{\mu}{\mu + \mathrm{diag}^{-1}(\Phi\Phi^\top)},\end{aligned} \tag{15}$$

where $\mathrm{diag}^{-1}(\cdot)$ denotes extracting the diagonal elements to form a vector.

By plugging Eq. (13), Eq. (14), Eq. (15) into Eq. (12) and simplifing the formula, we have

$$x^{k+1} = z^k + \Phi^\top\left(\frac{y - \Phi z^k}{\mu + \mathrm{diag}^{-1}(\Phi\Phi^\top)}\right). \tag{16}$$

However, in the DCCHI system, the dual-camera structure disrupts the diagonal property of $\Phi\Phi^\top$, as illustrated in Fig. 1, making it difficult to solve the equation in closed-form.

Eq. (11b) can be estimated with a proximal operator:

$$z^{k+1} = \mathrm{prox}_J(x^{k+1}). \tag{17}$$

Other algorithms that rely on a closed-form solution, such as the ADMM-based and GAP-based methods, have a similar derivation and encounter the same issue.

B. Overall Architecture

The dual-camera structure of DCCHI disrupts the diagonal property of the product of the sensing matrix and its transpose, as illustrated in Fig. 1, hindering the application of model-based methods and DUNs that rely on a closed-form solution. To address this issue, we propose an Alternating Direction DUN, as depicted in Fig. 3, which decouples the imaging model of DCCHI into a CASSI subproblem and a PAN subproblem while sharing a joint prior.

The proposed method is formulated as a nested optimization problem. It consists of an outer optimization loop that alternates between solving the CASSI subproblem and the PAN subproblem. Both subproblems are solved using optimization techniques and share a joint prior. Simplifying this process, we set the inner optimization iteration to 1, transforming it into an alternating solution approach for the CASSI data subproblem, the PAN data subproblem, and the joint prior subproblem.

By unfolding the proposed optimization method into a DUN and converting it into an RNN through parameter sharing across stages, the proposed ADRNN is obtained. Furthermore, to address the joint prior subproblem, we propose a Cross Spectral Transformer (XST), which effectively exploits the correlation between the PAN image and the compressed HSI.

C. Alternating Direction Recurrent Neural Network

Previous works [16], [36] have shown that a DUN can be treated as a Recurrent Neural Network (RNN). Therefore, we convert the DUN into an RNN by sharing parameters across stages, deriving the proposed ADRNN.

Overall, the proposed ADRNN is a nested HQS optimization algorithm. ADRNN starts from the maximum a posteriori (MAP) theory. In particular, the original HSI signal could be estimated by minimizing the following energy function as

$$\hat{x} = \arg\min_x \frac{1}{2}\|y_c - \Phi_c x\|_2^2 + \frac{1}{2}\|y_p - \Phi_p x\|_2^2 + J(x). \tag{18}$$

It can be rewritten in the following form:

$$\begin{aligned}\hat{x} &= \arg\min_x \frac{1}{2}\|y_c - \Phi_c x\|_2^2 + \lambda_c J(x) + \frac{1}{2}\|y_p - \Phi_p x\|_2^2 + \lambda_p J(x) \\ &= \arg\min_x f(x) + g(x),\end{aligned} \tag{19}$$

where λ_c and λ_p are balance parameters. By introducing an auxiliary variable z, Eq. (19) can be reformulated as

$$\hat{x} = \arg\min_x f(x) + g(z) \qquad s.t. \quad x = z, \tag{20}$$

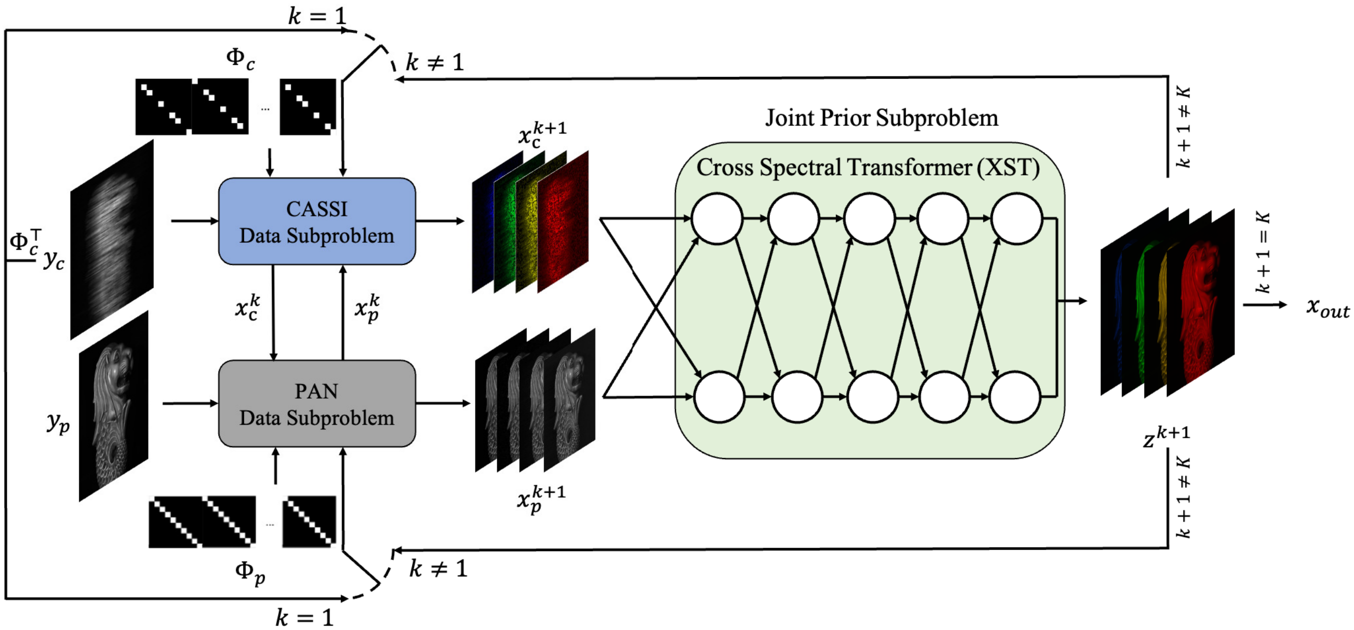

Fig. 3. The diagram of the Alternating Direction Recurrent Neural Network (ADRNN). The ADRNN solves the reconstruction problem of DCCHI by alternately solving a CASSI subproblem, a PAN subproblem, and a joint prior subproblem.

which is a constrained optimization problem. By adopting HQS algorithm, Eq. (20) is solved by minimizing

$$L(x, z) = f(x) + g(z) + \frac{\mu}{2}\|x - z\|_2^2, \tag{21}$$

where μ is a penalty parameter that forces x and z to approach the same fixed point. Subsequently, Eq. (21) can be solved by decoupling x and z into an x-subproblem (CASSI subproblem) and a z-subproblem (PAN subproblem) as

$$x^{k+1} = \arg\min_x f(x) + \frac{\mu}{2}\|x - z^k\|_2^2, \tag{22a}$$

$$z^{k+1} = \arg\min_z g(z) + \frac{\mu}{2}\|z - x^{k+1}\|_2^2. \tag{22b}$$

1) CASSI Subproblem: To prevent confusion and ensure simplicity, let us denote z^k simply as z and x as x_c in Eq. (22a). In this context, the CASSI subproblem can be represented as

$$\hat{x}_c = \arg\min_{x_c} \frac{1}{2}\|y_c - \Phi_c x_c\|_2^2 + \lambda_c J(x_c) + \frac{\mu}{2}\|x_c - z\|_2^2. \tag{23}$$

By introducing the auxiliary variable z_c, Eq. (23) can be reformulated as:

$$\hat{x}_c = \arg\min_{x_c} \frac{1}{2}\|y_c - \Phi_c x_c\|_2^2 + \lambda_c J(z_c) + \frac{\mu}{2}\|x_c - z\|_2^2 \quad s.t.\ x_c = z_c. \tag{24}$$

By adopting the HQS algorithm, the CASSI unfolding inference is obtained. Then, Eq. (24) is solved by minimizing

$$L_c(x_c, z_c) = \frac{1}{2}\|y_c - \Phi_c x_c\|_2^2 + \lambda_c J(z_c) + \frac{\mu}{2}\|x_c - z\|_2^2 + \frac{\mu_c}{2}\|x_c - z_c\|_2^2, \tag{25}$$

where μ_c is a penalty parameter that forces x_c and z_c to approach the same fixed point. Subsequently, Eq. (25) can be solved by decoupling x_c and z_c into an x_c-subproblem (CASSI data subproblem) and a z_c-subproblem (prior subproblem) as

$$x_c^{k+1} = \arg\min_{x_c} \frac{1}{2}\|y_c - \Phi_c x_c\|_2^2 + \frac{\mu}{2}\|x_c - z\|_2^2 + \frac{\mu_c}{2}\|x_c - z_c^k\|_2^2, \tag{26a}$$

$$z_c^{k+1} = \arg\min_{z_c} \lambda_c J(z_c) + \frac{\mu_c}{2}\|z_c - x_c^{k+1}\|_2^2. \tag{26b}$$

Eq. (26a) has a closed-form solution:

$$x_c^{k+1} = (\Phi_c^\top \Phi_c + (\mu + \mu_c)I)^{-1}(\Phi^\top y + \mu z + \mu_c z_c^k). \tag{27}$$

Following the matrix inverse lemma, the matrix inversion in Eq. (27) can be written as follows:

$$(\Phi_c^\top \Phi_c + (\mu + \mu_c)I)^{-1} = (\mu + \mu_c)^{-1} I - (\mu + \mu_c)^{-1}\Phi_c^\top (I + \Phi_c(\mu + \mu_c)^{-1}\Phi_c^\top)^{-1}\Phi_c(\mu + \mu_c)^{-1}. \tag{28}$$

In CASSI systems, $\Phi_c\Phi_c^\top$ is a diagonal matrix which can be defined as

$$\Phi_c\Phi_c^\top = \mathrm{diag}\{\phi_c^1, \ldots, \phi_c^i, \ldots, \phi_c^{HW'}\}, \tag{29}$$

where diag$\{\cdot\}$ denotes using the elements to form a diagonal matrix. By plugging $\Phi_c\Phi_c^\top$ into $(I + \Phi_c(\mu + \mu_c)^{-1}\Phi_c^\top)^{-1}$, we obtain:

$$\begin{aligned}(I + \Phi_c(\mu + \mu_c)^{-1}\Phi_c^\top)^{-1} &= \mathrm{diag}\left\{\frac{\mu + \mu_c}{(\mu + \mu_c) + \phi_c^1}, \ldots, \frac{\mu + \mu_c}{(\mu + \mu_c) + \phi_c^{HW'}}\right\} \\ &= \frac{\mu + \mu_c}{(\mu + \mu_c) + \mathrm{diag}^{-1}(\Phi_c\Phi_c^\top)},\end{aligned} \tag{30}$$

where diag$^{-1}(\cdot)$ denotes extracting the diagonal elements to form a vector. By plugging Eq. (28), Eq. (29), Eq. (30) into Eq. (27), we obtain:

$$x_c^{k+1} = \frac{\mu}{\mu + \mu_c} z + \frac{\mu_c}{\mu + \mu_c} z_c^k$$

$$+\frac{\mu}{\mu+\mu_c}\Phi_c^\top\left(\frac{y_c-\Phi_c z}{(\mu+\mu_c)+\mathrm{diag}^{-1}(\Phi_c\Phi_c^\top)}\right)$$
$$+\frac{\mu_c}{\mu+\mu_c}\Phi_c^\top\left(\frac{y_c-\Phi_c z_c}{(\mu+\mu_c)+\mathrm{diag}^{-1}(\Phi_c\Phi_c^\top)}\right). \tag{31}$$

The prior subproblem in Eq. (26b) can be estimated using a proximal operator:

$$z_c^{k+1} = \mathrm{prox}_J(x_c^{k+1}). \tag{32}$$

2) PAN Subproblem: To prevent confusion and ensure simplicity, let us denote x^{k+1} simply as x and z as x_p in Eq. (22b). In this context, the PAN subproblem can be represented as

$$\hat{x}_p = \arg\min_{x_p}\frac{1}{2}\|y_p-\Phi_p x_p\|_2^2+\lambda_p J(x_p)+\frac{\mu}{2}\|x_p-x\|_2^2. \tag{33}$$

Similarly, by introducing the auxiliary variable z_p, Eq. (33) can be reformulated as:

$$\hat{x}_p = \arg\min_{x_p}\frac{1}{2}\|y_p-\Phi_p x_p\|_2^2+\lambda_p J(z_p)+\frac{\mu}{2}\|x_p-x\|_2^2 \quad s.t.\ x_p = z_p. \tag{34}$$

To obtain the PAN unfolding inference, we adopt the HQS algorithm. Then, Eq. (34) is solved by minimizing

$$\hat{x}_p = \arg\min_{x_p}\frac{1}{2}\|y_p-\Phi_p x_p\|_2^2+\lambda_p J(z_p) + \frac{\mu}{2}\|x_p-x\|_2^2+\frac{\mu_p}{2}\|x_p-z_p\|_2^2, \tag{35}$$

where μ_p is a penalty parameter that forces x_p and z_p to approach the same fixed point. Subsequently, Eq. (35) can be solved by decoupling x_p and z_o into an x_p-subproblem (PAN data subproblem) and a z_p-subproblem (prior subproblem) as

$$x_p^{k+1} = \arg\min_{x_p}\frac{1}{2}\|y_p-\Phi_p x_p\|_2^2 + \frac{\mu}{2}\|x_p-x\|_2^2 + \frac{\mu_p}{2}\|x_p-z_p^k\|_2^2, \tag{36a}$$
$$z_p^{k+1} = \arg\min_{z_p}\lambda_p J(z_p)+\frac{\mu_p}{2}\|z_p-x_p^{k+1}\|_2^2. \tag{36b}$$

Similar to the CASSI subproblem, Eq. (36a) has a closed-form solution:

$$x_p^{k+1} = \frac{\mu}{\mu+\mu_p}x+\frac{\mu_p}{\mu+\mu_p}z_p^k + \frac{\mu}{\mu+\mu_p}\Phi_p^\top\left(\frac{y_p-\Phi_p x}{(\mu+\mu_p)+\mathrm{diag}^{-1}(\Phi_p\Phi_p^\top)}\right) + \frac{\mu_p}{\mu+\mu_p}\Phi_p^\top\left(\frac{y_p-\Phi_p z_p^k}{(\mu+\mu_p)+\mathrm{diag}^{-1}(\Phi_p\Phi_p^\top)}\right). \tag{37}$$

Also, the prior subproblem in Eq. (36b) can be estimated using a proximal operator:

$$z_p^{k+1} = \mathrm{prox}_J(x_p^{k+1}) \tag{38}$$

For simplicity, we set the number of iterations for the CASSI subproblem and PAN subproblem to 1, and let $\mu_c = \mu_p$, $\lambda_c = \lambda_p = 1$. From Eq. (18), we can observe that the CASSI sub-problem and the PAN sub-problem share a joint prior, and previous works [7], [16], [17], [35], [58], [59] have shown that a proximal operator can be implemented through a DNN. Thus, we propose a Cross Spectral Transformer (XST) to solve the joint prior subproblem, which will be detailedly illustrated in the next section. Therefore, the proposed ADRNN updates as follows:

Algorithm 1 Alternating Direction Recurrent Neural Network

Initialization:

(1) Set sensing matrices Φ_c, Φ_p, $\mu > 0$, $\mu_c = \mu_p > 0$, $\lambda_c = \lambda_p > 0$, $k = 1$;
(2) Set measurements y_c, y_p;
(3) Initialize $x_c^1 = x_p^1 = \Phi_c^\top y_c$;

While k $\leq$ K **do**

(1) Compute
$$x_c^{k+1} = \frac{\mu}{\mu+\mu_c}x_p^k+\frac{\mu_c}{\mu+\mu_c}z^k + \frac{\mu}{\mu+\mu_c}\Phi_c^\top\left(\frac{y_c-\Phi_c x_p^k}{(\mu+\mu_c)+\mathrm{diag}^{-1}(\Phi_c\Phi_c^\top)}\right) + \frac{\mu_c}{\mu+\mu_c}\Phi_c^\top\left(\frac{y_c-\Phi_c z^k}{(\mu+\mu_c)+\mathrm{diag}^{-1}(\Phi_c\Phi_c^\top)}\right);$$

(2) Compute
$$x_p^{k+1} = \frac{\mu}{\mu+\mu_p}x_c^k+\frac{\mu_p}{\mu+\mu_p}z^k + \frac{\mu}{\mu+\mu_p}\Phi_p^\top\left(\frac{y_p-\Phi_p x_c^k}{(\mu+\mu_p)+\mathrm{diag}^{-1}(\Phi_p\Phi_p^\top)}\right) + \frac{\mu_p}{\mu+\mu_p}\Phi_p^\top\left(\frac{y_p-\Phi_p z^k}{(\mu+\mu_p)+\mathrm{diag}^{-1}(\Phi_p\Phi_p^\top)}\right);$$

(3) Compute $z^{k+1} = \mathrm{CrossSpectralTransformer}(x_c^{k+1}, x_p^{k+1})$;
(4) $k = k+1$.

End while

Output: z^k

$$x_c^{k+1} = \frac{\mu}{\mu+\mu_c}x_p^k+\frac{\mu_c}{\mu+\mu_c}z^k + \frac{\mu}{\mu+\mu_c}\Phi_c^\top\left(\frac{y_c-\Phi_c x_p^k}{(\mu+\mu_c)+\mathrm{diag}^{-1}(\Phi_c\Phi_c^\top)}\right) + \frac{\mu_c}{\mu+\mu_c}\Phi_c^\top\left(\frac{y_c-\Phi_c z^k}{(\mu+\mu_c)+\mathrm{diag}^{-1}(\Phi_c\Phi_c^\top)}\right) \tag{39a}$$
$$x_p^{k+1} = \frac{\mu}{\mu+\mu_p}x_c^k+\frac{\mu_p}{\mu+\mu_p}z^k + \frac{\mu}{\mu+\mu_p}\Phi_p^\top\left(\frac{y_p-\Phi_p x_c^k}{(\mu+\mu_p)+\mathrm{diag}^{-1}(\Phi_p\Phi_p^\top)}\right) + \frac{\mu_p}{\mu+\mu_p}\Phi_p^\top\left(\frac{y_p-\Phi_p z^k}{(\mu+\mu_p)+\mathrm{diag}^{-1}(\Phi_p\Phi_p^\top)}\right) \tag{39b}$$
$$z^{k+1} = \mathrm{CrossSpectralTransformer}(x_c^{k+1}, x_p^{k+1}) \tag{39c}$$

In summary, the proposed iterative algorithm is detailed in Algorithm 1, where we initialize $x_c^1 = x_p^1 = \Phi_c^\top y_c$.

D. Cross Spectral Transformer

Previous methods [11], [20], [49], [50] have primarily focused on utilizing the PAN image to enhance the spatial information of the compressed HSI, yet they have neglected to utilize the compressed HSI to reciprocally enrich the spectral

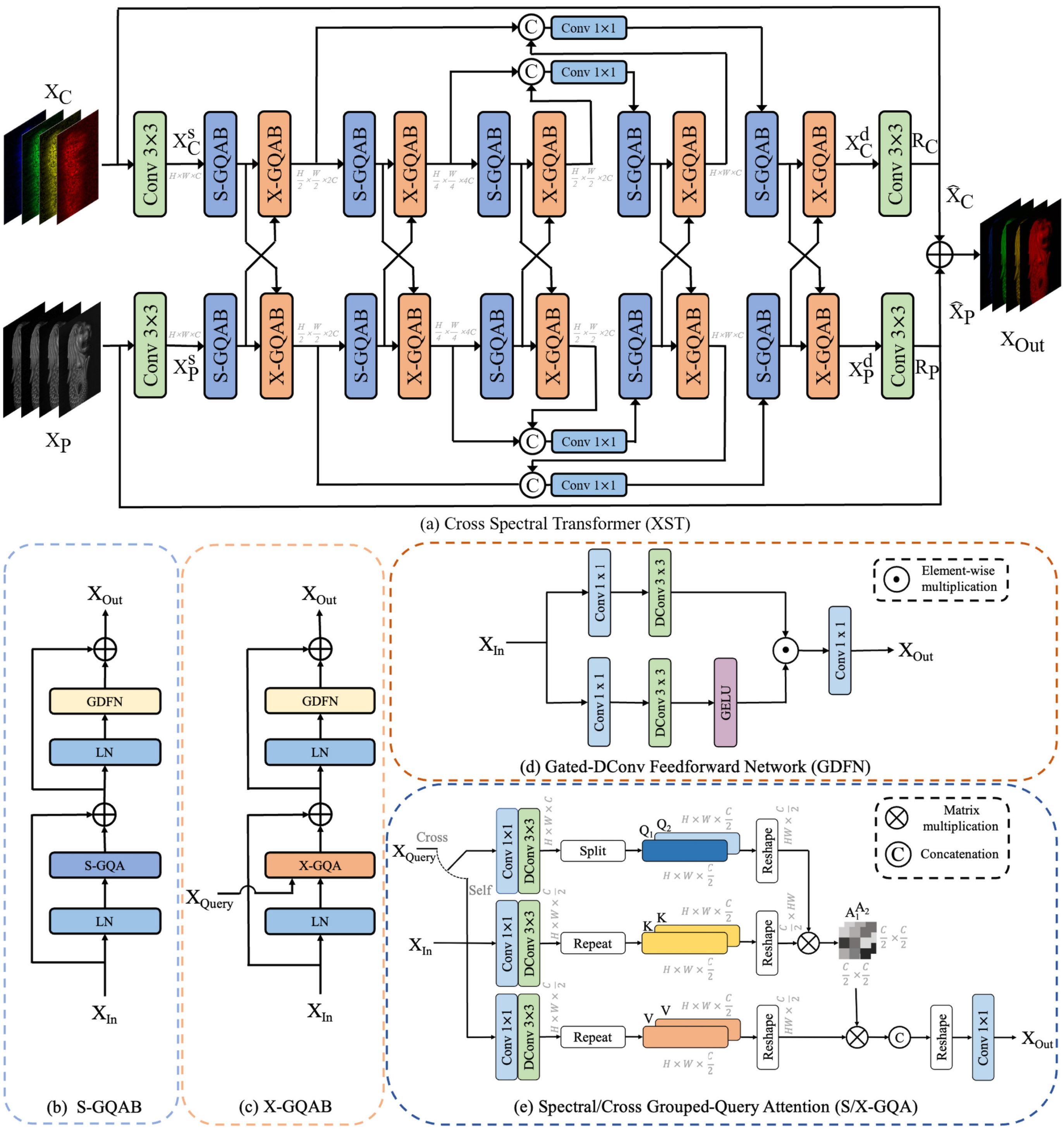

Fig. 4. The overall architecture of XST. (a) The diagram of XST. (b) S-GQAB is composed of two layer normalizations (LNs), a Spectral Grouped-Query Attention (S-GQA), and a Gated-DConv Feedforward Network (GDFN). (c) X-GQAB consists of two LNs, a Cross Grouped-Query Attention (X-GQA), and a GDFN. (d) The diagram of GDFN. (e) The diagram of S/X-GQA.

information of the PAN image. To address these issues, we propose a cross spectral attention mechanism to exploit the correlation between the HSI and the PAN image, treating the compressed HSI and the PAN image impartially. Additionally, we introduce the grouped-query attention (GQA) [2] to alleviate the burden of parameters and computational cost brought by impartially treating the compressed HSI and the PAN image.

As shown in Fig. 4 (a), XST adopts a dual-branch structure with triple up-down sampling and cross-interaction between branches. XST is constructed alternatively using basic units: Spectral Grouped-Query Attention Block (S-GQAB) and Cross Grouped-Query Attention Block (X-GQAB). Firstly, XST uses a Conv 3×3 to map the output of the CASSI data subproblem $\mathrm{X_C} \in \mathbb{R}^{H \times W \times N_\lambda}$ and the output of the PAN data subproblem $\mathrm{X_P} \in \mathbb{R}^{H \times W \times N_\lambda}$ into shallow features

$X_C^s \in \mathbb{R}^{H\times W\times C}$ and $X_P^s \in \mathbb{R}^{H\times W\times C}$, respectively. Then, these shallow features X_C^s and X_P^s pass through all S-GQABs and X-GQABs alternately, undergo three down and up samples, and are transformed into deep features $X_C^d \in \mathbb{R}^{H\times W\times C}$ and $X_P^d \in \mathbb{R}^{H\times W\times C}$ respectively. For feature downsampling and upsampling, we apply a Conv 3×3 with a stride of 2. To assist the recovery process, the features with the same scale are concatenated via skip connections. The concatenation operation is followed by a Conv 1×1 to reduce channels (by half) at all scales. Finally, a Conv 3×3 operates on deep features X_C^d and X_P^d to generate residual images $R_C \in \mathbb{R}^{H\times W\times N_\lambda}$ and $R_P \in \mathbb{R}^{H\times W\times N_\lambda}$ separately. The restoration results for the two branches, $\hat{X}_C$ and $\hat{X}_P$, are obtained by summing the input images, X_C and X_P, with the respective residual images, R_C and R_P. The final restored image, X_{Out}, is then achieved by summing $\hat{X}_C$ and $\hat{X}_P$.

1) Spectral Grouped-Query Attention Block (S-GQAB): As illustrated in Fig. 4 (b), the S-GQAB is composed of two layer normalizations (LNs), a Spectral Grouped-Query Attention (S-GQA), and a Gated-DConv Feedforward Network (GDFN) [57]. The most important component of S-GQAB is the proposed S-GQA module. S-GQA is inspired by S-MSA [6], [57]. When the pointer in Fig. 4 (e) points to the bottom, it demonstrates the diagram of S-GQA. GQA [2] divides heads into several groups and shares the key and value in each group. For simplicity, we illustrate the GQA with two heads and one group. The input tokens of S-GQA are denoted as $X_{In} \in \mathbb{R}^{H\times W\times C}$. Subsequently, the S-GQA generates *query*Q $\in \mathbb{R}^{H\times W\times C}$, *key*K $\in \mathbb{R}^{H\times W\times \frac{C}{2}}$, and *value*V $\in \mathbb{R}^{H\times W\times \frac{C}{2}}$, all enriched with local context. This is achieved by applying a Conv 1×1 to aggregate pixel-wise cross-channel context, followed by a DConv 3×3 to encode channel-wise spatial context, yielding

$$Q = W_d^Q W_p^Q X_{In}, K = W_d^K W_p^K X_{In}, V = W_d^V W_p^V X_{In}, \tag{40}$$

where $W_p^{(\cdot)}$ is the Conv 1×1 and $W_d^{(\cdot)}$ is the DConv 3×3, DConv denotes the depth-wise convolution [21]. Then, the *query*Q is split into two heads $Q_1, Q_2 \in \mathbb{R}^{H\times W\times \frac{C}{2}}$, and the *key*K, *value*V are repeated twice to match the number of the heads in each group. Next, the *query*, *key*, and *value* are reshaped to calculate attention. The S-GQA treats each spectral representation as a token and calculates the self-attention for each head$_i$:

$$A_i = \text{softmax}(\sigma_i K^\top Q_i), \quad \text{head}_i = VA_i, \quad i = 1,2, \tag{41}$$

where $K^\top$ denotes the transposed matrix of K. As the spectral density varies significantly with respect to the wavelengths, a learnable scaling parameter σ_i is used to reweight the dot product of $K^\top$ and Q before applying the softmax function. Subsequently, the outputs of all heads are concatenated spectrally to undergo a linear projection to generate the output X_{Out}:

$$X_{Out} = \text{S-GQA}(X_{In}) = W_p\text{Concat}(\text{head}_i) \quad i = 1,2. \tag{42}$$

2) Cross Grouped-Query Attention Block (X-GQAB): As shown in Fig. 4 (c), the X-GQAB consists of two layer normalizations (LNs), a Cross Grouped-Query Attention (X-GQA), and a GDFN [57]. The most important element of X-GQAB is the proposed X-GQA module. When the pointer in Fig. 4 (e) points to the left, it demonstrates the diagram of X-GQA. The X-GQA, also derived from the S-MSA [6], [57], primarily differs from the S-GQA in terms of the source of the query. The inputs of X-GQA are denoted as $X_{In} \in \mathbb{R}^{H\times W\times C}$ and $X_{Query} \in \mathbb{R}^{H\times W\times C}$. Subsequently, the X_{Query} is used to generate *query*Q $\in \mathbb{R}^{H\times W\times C}$, and the X_{In} is used to generate *key*K $\in \mathbb{R}^{H\times W\times \frac{C}{2}}$ and *value*V $\in \mathbb{R}^{H\times W\times \frac{C}{2}}$, all enriched with local context:

$$Q = W_d^Q W_p^Q X_{Query}, K = W_d^K W_p^K X_{In}, V = W_d^V W_p^V X_{In}. \tag{43}$$

Then, the *query*Q is split into two heads $Q_1, Q_2 \in \mathbb{R}^{H\times W\times \frac{C}{2}}$, and the *key*K, *value*V are repeated twice to match the number of the heads in each group. Next, the *query*, *key*, and *value* are reshaped to calculate attention. The X-GQA treats each spectral representation as a token and calculates the self-attention for each head$_i$:

$$A_i = \text{softmax}(\sigma_i K^\top Q_i), \quad \text{head}_i = VA_i, \quad i = 1,2. \tag{44}$$

Subsequently, the outputs of all heads are concatenated spectrally to undergo a linear projection:

$$X_{Out} = \text{X-GQA}(X_{In}, X_{Query}) = W_p\text{Concat}(\text{head}_i) \quad i = 1,2. \tag{45}$$

E. Analysis of Computational Complexity

1) Computational Complexity of ADRNN: When $\Phi\Phi^T$ is a non-diagonal matrix, the computational complexity of calculating the inverse $(I+\Phi\mu^{-1}\Phi^\top)^{-1}$ is $O(n^3)$, where n denotes the number of pixels in the HSI. Since Φ is a large and fat matrix, it is difficult to compute the inverse matrix [15], [46]. However, when $\Phi\Phi^T$ is a diagonal matrix, the computational complexity reduces to $O(n)$. Therefore, by decoupling the sensing matrix, we exploit the diagonal structure of both the $\Phi_c\Phi_c^T$ in the CASSI system and the $\Phi_p\Phi_p^T$ in the PAN system, thereby reducing the overall computational complexity to $O(2n)$.

2) Computational Complexity of GQA: We analyzed the computational complexity of S/X-GQA and compared it with S-MSA. Since the queries of GQA in each group share a key and a value, the computational complexity of key and value projection can be reduced. The computational complexity analysis of S/X-GQA and S-MSA with two heads is as follows in Eq. (46):

$$\begin{aligned} \text{O(S/X-GQA)} &= \underbrace{HWC^2}_{\text{Q}} + \underbrace{HW\frac{C^2}{2}}_{\text{K and V}} + \underbrace{\frac{HWC^2}{2}}_{\text{A}} + \underbrace{\frac{HWC^2}{2}}_{\text{Out}} \\ &= 2HWC^2 + HW\frac{C^2}{2} \\ \text{O(S-MSA)} &= \underbrace{HWC^2}_{\text{Q}} + \underbrace{2HWC^2}_{\text{K and V}} + \underbrace{\frac{HWC^2}{2}}_{\text{A}} + \underbrace{\frac{HWC^2}{2}}_{\text{Out}} \\ &= 4HWC^2 \end{aligned} \tag{46}$$

Compared to S-MSA [6], [57], the reduction in computational complexity of S/X-GQA mainly lies in the projection of the *key* and *value*. When computing self/cross attention, the key and value need to be repeated to match the dimensions of the query, thus not reducing the computational complexity of self/cross attention. The GQA compresses the knowledge in MSA into fewer parameters, thus the projection features of the *key* and *value* in GQA are more representative.

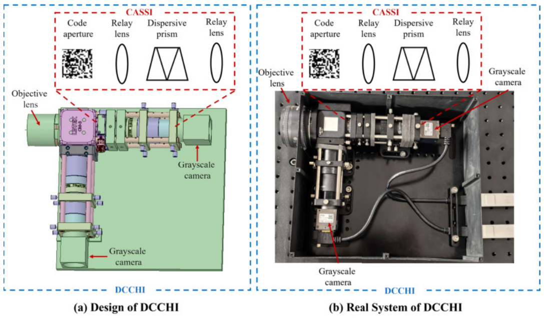

Fig. 5. The design and DCCHI and the real system of DCCHI.

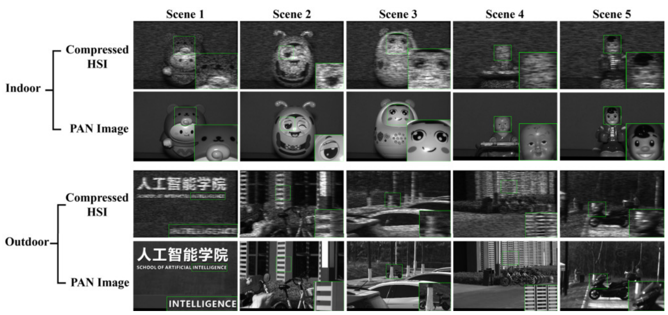

Fig. 6. Demonstration of real measurement pairs for indoor scenes and outdoor scenes. A patch is enlarged in the bottom-right corner for a better view of the spectral dispersion.

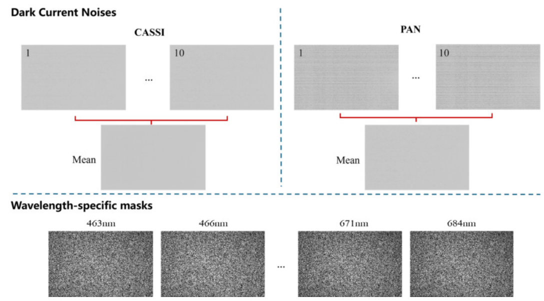

Fig. 7. The dark current noises and the wavelength-specific masks for system calibration.

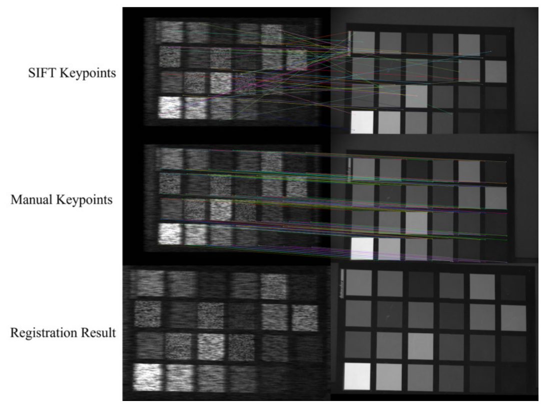

Fig. 8. Automatically detected keypoints matching results, manually annotated keypoints matching results, and the registration result using manually annotated keypoints.

V. Experiments

A. Simulation Setting

In simulation experiments, 28 wavelengths are selected from 450nm to 650nm and derived by spectral interpolation for the HSI data. We utilize two datasets for our simulation experiments, specifically the CAVE [54] and KAIST [12] datasets. The compared models were all trained on the CAVE dataset and subsequently tested on the KAIST dataset. For fair comparisons, we selected 10 scenes of a spatial size 256×256 from the KAIST dataset for testing, following previous works [6], [16], [17], [22], [23], [31], [59].

To quantitatively evaluate the proposed method and the comparison methods, we adopt Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [51], and Spectral Angle Mapper (SAM) [25] as quantitative evaluation metrics. The SAM is calculated as follows:

$$\mathrm{SAM}(\mathbf{a}, \mathbf{b}) = \cos^{-1}\left(\frac{\mathbf{a} \cdot \mathbf{b}}{||\mathbf{a}|| \cdot ||\mathbf{b}||}\right) \tag{47}$$

B. Real System and Setting

1) Real System: The design and the real DCCHI system are shown in Fig. 5. The prism produces a horizontal dispersion of 39 pixels, corresponding to 40 spectral channels ranging from 463 nm to 684 nm. We have displayed the captured indoor and outdoor scenes in Fig. 6. The spatial resolution of the indoor scenes and outdoor scenes are 1024×1536.

2) Dark Current Noises: To reduce dark current noise in DCCHI imaging, we subtracted the mean dark current from the measurements. Dark current noise was calculated by averaging 10 dark frames per branch. As shown in the top portion of Fig. 7, the CASSI branch exhibits lower dark current noise compared to the PAN branch. We suggest such difference may be attributed to variations in detector characteristics or the suppression of structured noise by the coded aperture. While dark current is primarily thermally induced, partial light blockage may reduce photo-induced dark current components, contributing to the observed noise reduction.

3) Wavelength-Specific Masks: The sensing matrix is the core of the DCCHI system and is highly related to the physical masks. The number of dispersed pixels determines how many spectral bands the system can reconstruct, while the integer pixels at specific wavelengths define the corresponding reconstructible wavelengths. By capturing masks at various wavelengths, the system's sensing matrix can be derived. The bottom portion of Fig. 7 shows some of the captured masks.

4) Dual-Camera Registration: To achieve registration between the two branches of the DCCHI system, we matched keypoints between images from the two branches. This process allowed us to calculate the affine transformation matrix and then apply the affine transformation to the PAN image. We used the color checker scene to derive the keypoints because of its distinctive and easily identifiable keypoints. We experimented with both automatically detected keypoints and manually annotated keypoints. The top part of Fig. 8 shows the matching results of keypoints detected by the SIFT operator. The matching performance was poor, making automated registration challenging. Given the distinctiveness of the color checker's keypoints, we manually annotated 96 keypoints from images of the two branches in the color checker scene. The matching results for these manually annotated keypoints are presented in the middle part of Fig. 8. Using the manually annotated keypoints, we calculated the affine

TABLE I

COMPARISONS BETWEEN ADRNN-XST AND SOTA TECHNIQUES ACROSS 10 SIMULATION SCENES. THE PRESENTED RESULTS INCLUDE PSNR (TOP ENTRY OF EACH CELL), SSIM (MIDDLE ENTRY OF EACH CELL) AND SAM (BOTTOM ENTRY OF EACH CELL)

Algorithms	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Avg
TwIST [4]	31.68 0.854 17.185	27.38 0.712 28.469	20.34 0.431 31.166	24.81 0.699 26.899	28.77 0.778 16.892	29.26 0.776 22.498	25.07 0.646 24.866	27.89 0.774 26.280	22.07 0.623 25.946	27.53 0.730 26.454	26.48 0.702 24.666
GAP-TV [55]	32.85 0.892 16.793	28.49 0.770 28.477	21.83 0.534 30.514	26.22 0.756 26.437	30.37 0.865 17.040	30.81 0.831 24.799	26.20 0.736 24.521	28.63 0.802 27.027	22.97 0.707 24.701	27.54 0.742 28.231	27.59 0.764 24.854
DeSCI [27]	28.48 0.769 29.024	22.79 0.712 24.661	29.65 0.820 26.904	26.70 0.774 23.273	32.21 0.851 24.294	31.31 0.892 16.676	26.15 0.752 26.590	21.43 0.529 31.050	28.62 0.792 27.560	33.63 0.902 16.223	28.12 0.779 24.626
λ-Net [34]	34.95 0.949 10.753	36.71 0.957 12.610	34.44 0.948 8.418	41.17 0.974 12.205	33.85 0.963 8.305	35.53 0.962 11.516	34.25 0.933 8.647	31.87 0.945 13.887	30.75 0.935 11.496	32.78 0.948 11.894	34.63 0.951 10.973
DGSMP [22]	36.12 0.964 9.190	36.51 0.965 11.425	32.77 0.936 8.266	39.19 0.960 13.179	34.73 0.972 7.698	37.48 0.975 10.240	35.12 0.946 7.414	35.44 0.971 11.510	32.46 0.956 9.529	35.93 0.978 10.155	35.57 0.962 9.861
PIDS [11]	39.82 0.977 6.705	37.07 0.921 13.121	37.72 0.950 6.356	46.78 0.978 10.907	37.45 0.973 6.292	37.74 0.963 13.050	32.90 0.896 8.341	31.66 0.915 15.994	34.35 0.902 8.629	38.58 0.972 11.202	37.41 0.945 10.060
MST [6]	37.82 0.974 7.141	40.61 0.983 8.734	38.09 0.970 5.311	44.13 0.986 9.060	37.36 0.981 5.931	38.98 0.983 8.837	36.68 0.957 6.205	36.77 0.979 10.052	36.96 0.974 7.213	37.83 0.984 9.078	38.52 0.977 7.756
DAUHST 9stg [7]	40.95 0.983 5.394	45.14 0.990 6.779	43.00 0.981 3.955	47.99 0.990 6.682	39.88 0.986 4.332	41.10 0.988 7.431	39.99 0.976 4.611	40.09 0.986 8.731	42.52 0.986 4.908	39.84 0.989 7.363	42.05 0.985 6.019
RDLUF-MixS^2 9stg [17]	41.75 0.986 5.016	46.74 0.993 5.922	44.75 0.984 3.466	49.46 0.994 6.246	40.54 0.989 4.051	41.78 0.990 6.670	41.19 0.979 4.247	40.75 0.988 7.302	44.76 0.989 4.236	40.82 0.990 6.057	43.25 0.988 5.321
In2SET 9stg [50]	42.59 0.987 4.932	46.39 0.992 6.676	44.49 0.983 3.892	50.36 0.993 8.599	42.01 0.990 4.238	42.43 0.988 8.642	41.62 0.981 4.041	40.57 0.986 9.388	43.86 0.987 5.359	42.36 0.991 7.739	43.67 0.988 6.351
DERNN-LNLT 9stg [16]	42.83 0.987 4.469	48.15 0.994 5.443	45.04 0.984 3.187	51.43 **0.996** 5.240	41.97 0.990 3.567	42.78 0.991 6.212	41.88 0.982 3.910	42.36 0.991 6.526	46.11 0.990 3.844	42.89 0.993 5.286	44.54 0.990 4.768
ADRNN-XST 9stg	**43.46** **0.989** **4.193**	**49.45** **0.995** **4.882**	**45.85** **0.986** **2.962**	**51.9** **0.996** **5.058**	**42.86** **0.993** **3.095**	**43.73** **0.993** **5.631**	**42.99** **0.985** **3.687**	**43.37** **0.992** **5.901**	**46.51** **0.991** **3.658**	**43.56** **0.994** **4.991**	**45.37** **0.991** **4.406**

transformation matrix and applied it to the PAN image for registration with the compressed HSI. The registration result is shown in the bottom part of Fig. 8.

5) Real Setting: In real experiments, the compared models were trained on CAVE [54] and ICVL [3] datasets and tested on the real indoor scenes and outdoor scenes captured by our built DCCHI system. To simulate real measurement conditions, 11-bit shot noise was injected in the training measurements.

C. Quantitative Results

To showcase the effectiveness of the proposed method, we performed a comparative analysis with eleven established methods using simulation datasets. The methods mentioned encompass four traditional model-based techniques (TwIST [4], GAP-TV [55], DeSCI [27], PIDS [11]), two end-to-end neural networks methods (λ-Net [34] and MST [6]), and five DUNs (DGSMP [22], DAUHST [7], RDLUF-MixS^2 [17], In2SET [50], DERNN-LNLT [16]). The inputs for all CASSI-based methods are both the compressed HSI and the PAN image. For traditional model-based methods that solve the data subproblem using gradient descent, such as TwIST, we use the sensing matrix of the DCCHI system for gradient descent calculation. For traditional model-based methods that are unable to obtain a closed-form solution due to the disrupted diagonal property in DCCHI, such as GAP-TV and DeSCI, we also use gradient descent to solve the data subproblem similarly to TwIST. For deep learning-based methods [6], [7], [16], [17], [22], [34], we concatenate the PAN Image with the compressed HSI as additional input for end-to-end networks and the denoiser of DUNs. The results for 10 simulated scenes can be found in Table. I. End-to-end neural networks and DUNs have outperformed most traditional model-based methods. PIDS, with its superior prior image semantic similarity regularization, has surpassed both λ-Net and DGSMP. Both end-to-end neural networks and DUNs achieve excellent performance, showcasing the effectiveness of deep learning. The proposed method surpassed all compared methods in all metrics. Specifically, compared to λ-Net, DGSMP, PIDS, MST, DAUHST 9stg, RDLUF-MixS2 9stg, In2SET 9stg and DERNN-LNLT 9stg, our proposed ADRNN-XST 9stg outperformed them with an average improvement of 10.74 dB, 9.80 dB, 7.96dB, 6.85 dB, 3.32 dB, 2.12 dB, 1.70dB, and 0.83 dB, respectively.

The proposed method achieves the lowest SAM among all compared methods, demonstrating its superior ability in reconstructing the original spectral signals. Specifically, a lower SAM value implies the smallest spectral angle difference between the reconstructed spectrum and the ground truth spectrum, thereby minimizing the reconstruction error while maintaining consistency in spectral features. To further analyze the improvement of the proposed method over the comparison methods in terms of SAM, we plotted the spectral curves of the proposed method, comparison methods, and ground truth across multiple regions in simulation scene 7. As shown in

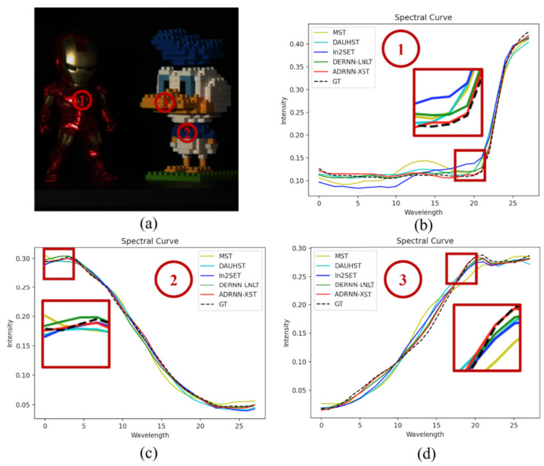

Fig. 9. Spectral curves of multiple regions in Simulation Scene 7 for different comparison methods.

Fig. 9, by comparing the spectral curves of different methods with the ground truth spectral curves, it can be observed that: the reconstructed spectral results of the proposed method exhibit the highest similarity and smallest deviation compared to the ground truth spectral curves.

D. Qualitative Results

1) Simulation Results: To conveniently assess the visual quality, we visualized error maps and spectral density curves of the reconstructed results generated by various SOTA techniques and the proposed ADRNN-XST, including two traditional model-based methods (GAP-TV [55] and PIDS [11]), one end-to-end neural networks (MST [6]), and four DUNs (DGSMP [22], [23], DAUHST [7], RDLUF-MixS^2 [17], In2SET [50]). The right of Fig. 10 presents the error heatmaps for selected spectral bands. The comparison highlights the impressive capability of our ADRNN-XST to preserve spatial fidelity. In contrast, previous methods either introduce undesired distortions and blotchy textures that are absent in the Ground Truth (GT) or yield over-smooth results, compromising fine-grained structures. This is not only attributed to the PAN data subproblem continuously deriving spatial information from the PAN image but also the XST exploiting cucial information from the PAN image with cross attention. The bottom-left of Fig. 10 shows the spectral density curves. The spectral curves of ADRNN-XST demonstrate the highest correlation and coincidence with the reference curves, which showcases the superiority of our ADRNN-XST in reconstructing spectral-dimension consistency.

2) Real Results: To validate the effectiveness of the proposed method on real scenes, as shown in Fig. 11 (a), we developed a validation system that could simultaneously capture measurement pairs from the DCCHI system while acquiring reference HSIs. The system also uses a beam splitter to separate the light, with one path going to the spectrometer and the other to the DCCHI system. We reconstructed the color checker scene and compared the spectral curves of the red and green patches with those from the reference HSI in Fig. 11 (b). As illustrated in Fig. 11 (c), the reconstructed spectral curves showed consistent trends with the reference spectral curves, indicating the generalization and effectiveness of the proposed method in real-world scenarios.

Additionally, we conducted experiments with two model-based methods (TwIST [4] and GAP-TV [55]), two end-to-end neural networks (λ-Net [34] and MST [6]), two DUNs (DAUHST [7] and RDLUF-MixS^2 [17]), and our ADRNN-XST. We've conducted a comparison of the reconstructed results from real indoor scene 4 and real outdoor scene 1. Fig. 12 illustrates the reconstruction results for four spectral bands in these scenes. In indoor scene 4, TwIST and GAP-TV exhibited significant occurrences of blur and noise in their reconstructions. Notably, MST produced numerous block artifacts, while DAUHST's performance was adversely affected, resulting in patchy textures and perceptible blurring effects. In outdoor scene 1, TwIST and GAP-TV demonstrated severe noise and blurring, which critically hindered the readability of text edges. While λ-Net achieved some reduction in noise, it suffered from over-smoothing issues, leading to a loss of fine textual details. RDLUF-MixS^2 showed improved text clarity compared to previous methods; however, it still exhibited noticeable text distortions in certain spectral bands. In contrast, our ADRNN-XST can reconstruct clearer content and detailed textures with fewer artifacts and less blurring. The comparative analysis underscores our model's exceptional capability in image restoration, thereby affirming its application viability in real-world scenarios.

E. Analysis of ADRNN

To analyze the roles of the ADRNN in-depth, we visualize the solution of the CASSI data subproblem x_c in Eq. (39a), the solution of the PAN data subproblem x_p in Eq. (39b), and the solution of the joint prior subproblem z in Eq. (39c), and plot the PSNR-Stages curves of x_k and x_p in Fig. 13. The specific PSNR of x_c, x_p, and z at different stages is demonstrated in Table II. From Fig. 13 and Table II, we observe that: 1) Solving the PAN data subproblem brings more improvement than solving the CASSI data subproblem until Stage 5 and solving the CASSI data subproblem catches up with the performance of solving the PAN data subproblem at Stage 5 and surpasses it afterwards. This is because in the early stages, the high resolution of PAN brought higher benefits, and when the spatial resolution is sufficient, the spectral information implicit in the compressed HSI brings more benefits. 2) The improvements brought by solving data subproblems are very significant in the first few iterations, while the improvements brought by solving the prior subproblem gradually surpass those from the data subproblem after Stage 4. Specifically, at Stage 2, solving the CASSI data subproblem brings an improvement of 4.51dB, and solving the PAN data subproblem brings an improvement of 7.48dB. In contrast, at Stage 5, solving the prior subproblem yields an improvement of 1.00 dB, while solving both the CASSI data subproblem and the PAN data subproblem each bring an improvement of 0.26 dB.

F. Model Settings and Additional Comparisons

The compared models include CASSI-based methods [4], [6], [7], [16], [17], [27], [34], [55] and DCCHI-based methods [11], [50]. The inputs for CASSI-based methods are both the compressed HSI and the PAN image. For model-based methods that solve the data subproblem using gradient descent, such as TwIST, we use the sensing matrix of the DCCHI system for gradient descent calculation. For model-based methods that cannot obtain a closed-form solution due to the disrupted diagonal property in DCCHI, such as GAP-TV and

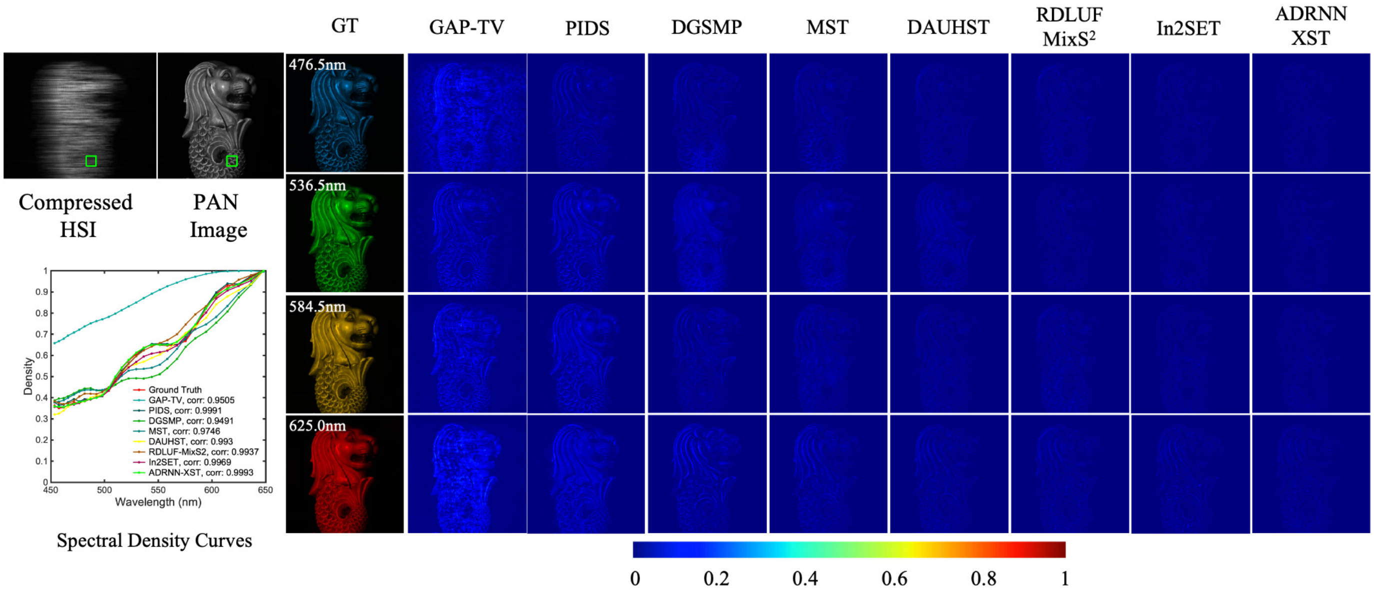

Fig. 10. Simulation Results of Scene 10. The top-left displays CASSI measurement (Compressed HSI) and the PAN measurement (PAN Image). The bottom-left shows the spectral curves corresponding to the two measurements marked with green boxes. The right presents comparative error maps of various SOTA methods at spectral bands 476.5nm, 536.5nm, 584.5nm, and 625.0nm.

TABLE II

THE PSNR OF x_c, x_p, AND z AT DIFFERENT STAGES

	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7	Stage 8	Stage 9
x_c	19.86	29.81 (+4.57)	37.12	40.99	42.75 (+0.26)	43.82	44.51	44.91	45.15
z	25.24	34.56	39.91	42.49	43.75 (+1.00)	44.45	44.85	45.10	45.37
x_p	25.66	32.72 (+7.48)	38.26	41.11	42.75 (+0.26)	43.79	44.48	44.89	45.13

TABLE III

COMPARISONS OF PARAMS, MEMORY COST, GFLOPS, TRAINING AND INFERENCE TIME

Method	Batch Size	Learning Rate	Training Time	Params	Memory	GFlops	Inference Time
TwIST	-	-	-	-	-	-	584s (CPU)
GAP-TV	-	-	-	-	-	-	874s (CPU)
λ-Net	5	1.5e-4	**1 day**	32.74M	84.42G	2096.8	38s (CPU)
MST	5	4e-4	3 days	4.04M	**7.62G**	**1270.4**	20s
DAUHST	5	4e-4	4 days	67.43M	17.98G	5708.6	17s
RDLUF-MixS^2	1	2e-4	4 days	11.52M	18.26G	4820.0	15s
DERNN-LNLT	1	3e-4	4 days	22.11M	18.36G	10877.9	16s
ADRNN-XST	1	3e-4	2 days	**2.69M**	12.85G	2724.8	**12s**

DeSCI, we similarly employ gradient descent to solve the data subproblem. For deep learning-based methods [6], [7], [16], [17], [22], [34], we concatenate the PAN image with the compressed HSI as additional input for end-to-end networks and the denoisers of DUNs.

In Tab. III, we provided detailed information about both the proposed method and the compared methods, including batch size, learning rate, training time, number of parameters, memory consumption, computational complexity, and inference time in real scenes (spatial size of 1024 × 1536). All hyperparameters were set based on the original paper or open-source code. All deep learning-based methods were trained on a server equipped with an 80GB Tesla A100 GPU, and all methods were inferred on another server with a 48GB Tesla L40 GPU, a 64-core, 128-thread Intel Xeon Gold 6326 CPU running at 2.90 GHz, and 512GB of RAM. As shown in Tab. III, the proposed method achieves the fewest parameters and the shortest inference time while maintaining relatively low memory consumption and computational cost. The results demonstrate that GFlops do not consistently correlate with inference time; despite MST's lower GFlops, its inference time is longer. This discrepancy is likely due to the heavy loop-based operations in computing mask-guided attention within MST, which may hinder GPU optimization. Additionally,

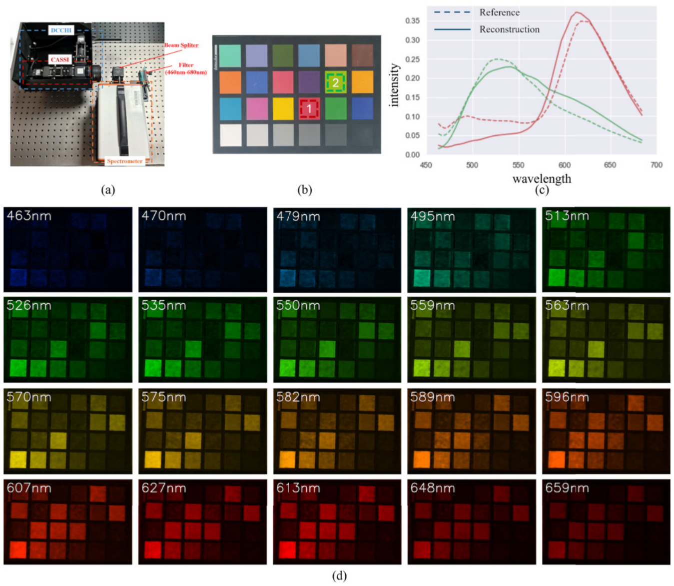

Fig. 11. The real reconstruction results of the color checker scene. (a) The real experimental validation system. (b) Color checker and the selected red and green patches for spectral curve comparison. (c) Spectral curves of the reconstructed HSI's red and green patches compared to those of the reference HSI's red and green patches. (d) The visualization of 20 out of 40 reconstructed spectral bands in the color checker scene.

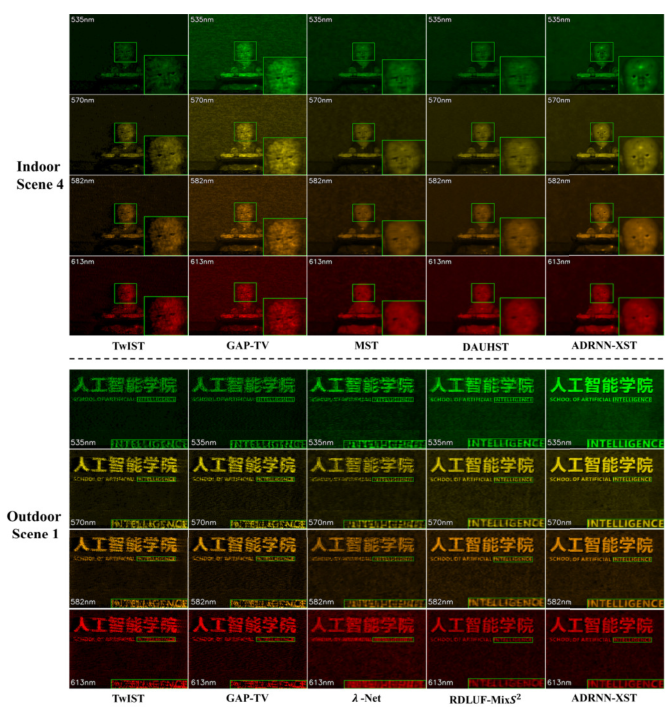

Fig. 12. Visual comparisons of various SOTA methods for indoor scene 4 and outdoor scene 1 captured from our DCCHI system at spectral bands 535nm, 570nm, 582nm, and 613nm.

λ-Net has high memory consumption due to its computation of global spatial attention maps, which becomes problematic at large scales, exceeding GPU capacity and requiring inference on the CPU. Furthermore, position encodings in DAUHST and DERNN-LNLT models introduce significant parameter overhead, particularly when processing high spatial resolutions. The fewest parameters, the shortest inference time, and relatively low memory consumption and computational cost further substantiate the superiority of the proposed method.

TABLE IV

Ablation Studies of Different Methods for Solving the Data Subproblem

	PSNR	SSIM	SAM
Gradient Descent	44.31	0.990	5.417
Close-form solution	44.65	0.990	4.821
Alternative Direction	45.37	0.991	4.406

TABLE V

Ablation Studies of Different Types of Attention Mechanisms

	Params	GFlops	PSNR	SSIM	SAM
MHA	1.36M	165.19	45.29	0.991	**4.406**
MQA	1.25M	158.40	45.17	0.991	4.497
GQA	1.28M	159.33	**45.37**	0.991	**4.406**

G. Ablation Study

In this section, we conducted ablation studies to validate the effectiveness of each element of the ADRNN-XST, using simulated HSI datasets [12], [54].

1) Gradient Descent Vs Closed-Form Solution Vs Alternating Direction: To validate the effectiveness of the proposed alternating direction mechanism, we conducted different methods to solve the data subproblem, including Gradient Descent (GD), Closed-form Solution (CS), and Alternating Direction (AD). The results are illustrated in Tab. IV. The closed-form solution achieves higher performance compared to gradient descent, suggesting the closed-form solution provides a more accurate solution compared to gradient descent. The proposed alternating direction mechanism outperforms the closed-form solution by 0.72dB, showcasing its effectiveness.

2) GQA Vs MQA Vs MHA: To validate the effectiveness of the Grouped-Query Attention (GQA), we replace the GQA in the XST with Multi-Head Attention (MHA) and Multi-Query Attention (MQA) respectively. The MQA divides the query into multiple heads, and all heads share one key and value. The ablation results are shown in Tab. V. The performance of MQA is reduced by 0.12dB compared to MHA, and the performance of GQA is 0.2dB higher than that of MQA. Interestingly, we observe that GQA achieves better results than MHA even with fewer Params and GFlops. This could suggest that some parameters in the MHA are redundant. Additionally, the shared key and value compress knowledge into limited parameters, thereby enhancing the representation capability of networks.

3) Cross-Attention Vs Self-Attention: To validate the effectiveness of the proposed XST, we replace it with a CNN-based architecture, U-Net [39], where the input is formed by concatenating the outputs from the CASSI data subproblem and the PAN data subproblem. To further evaluate the contribution of the Cross Grouped-Query Attention (X-GQA), we replace all X-GQABs with the S-GQAB in the XST, resulting in a variant without communication between the two branches. The results are demonstrated in Tab. VI. Benefiting from the more advanced and modern network architecture, even the prior learning network without inter-branch communication

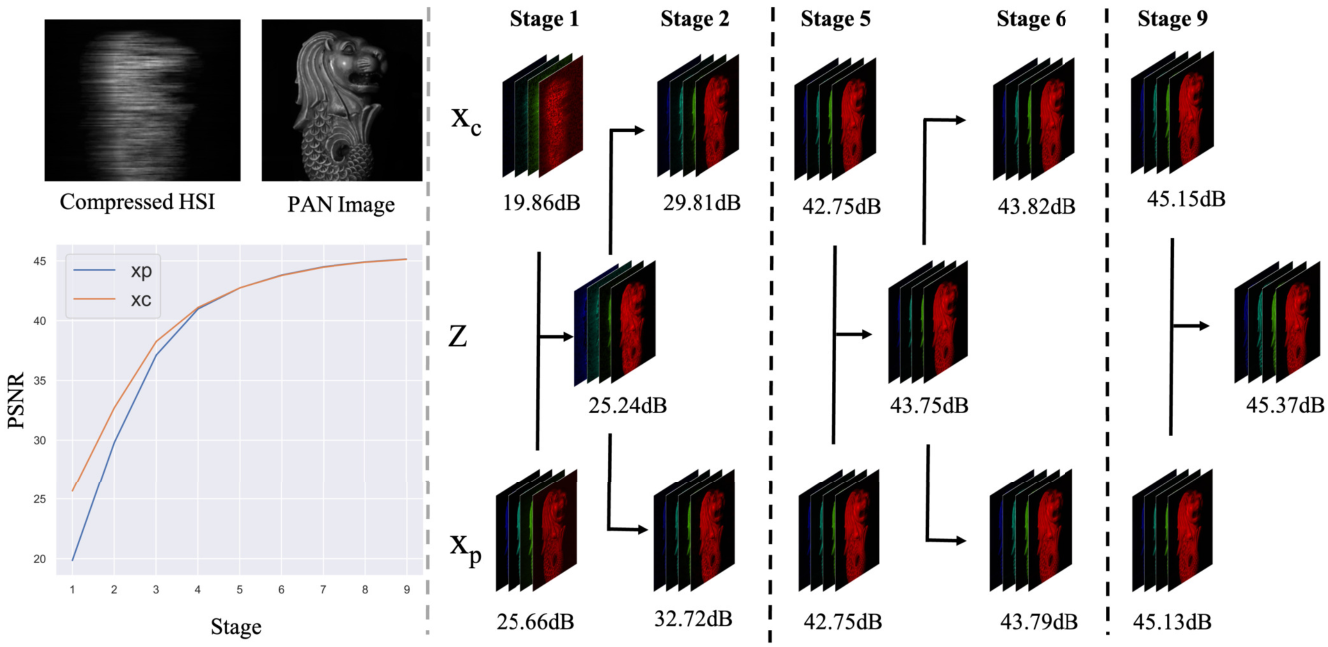

Fig. 13. Visualization of the solution of the CASSI data subproblem, x_c, the solution of the PAN data subproblem, x_p, and the solution of the joint prior subproblem, z, at different stages. The bottom-left corner depicts the PSNR-Stage curves for x_c and x_p.

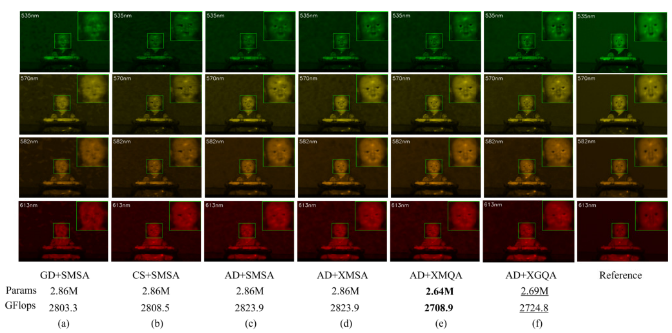

Fig. 14. Ablation studies in the real indoor scene 4.

TABLE VI
ABLATION STUDIES OF CROSS-ATTENTION MECHANISMS

	Params	GFlops	PSNR	SSIM	SAM
U-Net	31.07M	453.20	30.79	0.845	12.878
w/o X-GQA	1.28M	159.33	45.03	0.991	4.476
w X-GQA	1.28M	159.33	**45.37**	0.991	**4.406**

achieves significantly better performance than U-Net with much fewer parameters and computational cost. Additionally, we can observe that, with the same Params and GFlops, the XST with the X-GQA performs better than it does when all attentions are S-GQA. This demonstrates the effectiveness of X-GQA in exploiting the correlation between the PAN image and the compressed HSI.

4) Ablation Studies on the Real Scene: To further evaluate the effectiveness of ADRNN-XST, we conducted ablation studies on real indoor scene 4, with results shown in Fig. 14. Replacing the Alternating Direction (AD) mechanism with Gradient Descent (GD) or Closed-form Solution (CS), and substituting S-GQA/X-GQA with S-MSA, yielded baseline models GD-SMSA and CS-SMSA, which produced blurred and artifact-ridden reconstructions (Fig. 14 (a)–(b)). Incorporating AD to solve the data subproblem (AD-SMSA, Fig. 14 (c)) improved facial clarity, validating the AD mechanism's benefit. Replacing one S-MSA with X-MSA (AD-XMSA, Fig. 14 (d)) further enhanced detail reconstruction, highlighting X-MSA's ability to model correlations between HSI and PAN data. Finally, replacing MSA with MQA and GQA resulted in AD-XMQA (Fig. 14 (e)) and ADRNN-XST (AD-XGQA, Fig. 14 (f)), respectively. ADRNN-XST outperformed both AD-XMSA and AD-XMQA, particularly in reconstructing fine details such as the left eye at 613 nm. Despite fewer parameters and computational cost, GQA demonstrated superior performance over MHA, suggesting that MHA contains redundancy, while GQA enhances representation by compressing knowledge into fewer parameters.

TABLE VII
PERFORMANCE UNDER VARYING NOISE LEVELS

Noise Level	PSNR	SSIM	SAM
σ = 0	45.37	0.991	4.406
σ = 10	45.38	0.991	4.404
σ = 30	44.31	0.990	4.732
σ = 50	42.12	0.988	5.845
σ = 70	40.38	0.983	6.522

5) Performance Under Varying Noise Levels: To further assess the method's performance under varying noise levels, we conducted experiments with Gaussian noise variances of 10, 30, 50, and 70. Results are detailed in Tab. VII, where a variance of 0 corresponds to Poisson noise only. At a low noise level (variance =10), the proposed method shows a slight performance improvement, possibly due to noise acting as a kind of data augmentation, enhancing robustness and generalization. However, as noise increases, performance gradually degrades: PSNR decreases by 1.07 dB, 2.19 dB, and 1.74 dB; SSIM drops by 0.001, 0.002, and 0.005; and SAM increases by 0.328, 1.113, and 0.677 at variances of 30, 50, and 70, respectively. PSNR and SAM trends show an initial sharp decline followed by a slower decrease, indicating model's resilience to high noise. In contrast, SSIM continuously decreases with increasing noise levels, indicating that higher noise levels significantly disrupt the structural similarity of the images. Overall, the method demonstrates robustness across noise levels, benefiting slightly from low noise but suffering noticeable degradation under high noise.

VI. Conclusion

In this paper, we propose an alternating direction DUN, named ADRNN, which addresses the challenge of solving the data subproblem in closed-form in the DCCHI system by decoupling the imaging model of DCCHI into a CASSI subproblem and a PAN subproblem. In these subproblems, data terms and a joint prior are alternately solved. Additionally, a Cross Spectral Transformer (XST) is proposed to exploit the joint prior. The XST utilizes cross spectral attention to exploit the correlation between the compressed HSI and the PAN image, and incorporates GQA to alleviate the burden of parameters and computational cost brought by impartially treating the compressed HSI and the PAN image. Furthermore, we built a real DCCHI system and captured large-scale indoor and outdoor scenes for future academic research. The proposed ADRNN-XST achieved SOTA performance on both simulation and real datasets.

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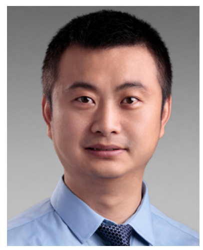
Yubo Dong (Member, IEEE) received the Ph.D. degree in computer science and technology from Xidian University, Xi'an, Shaanxi, China, in 2025.

He is currently a Postdoctoral Researcher at the School of Artificial Intelligence, Xidian University. His research interests include computational spectral imaging, image processing, and computer vision.



Dahua Gao (Member, IEEE) received the Ph.D. degree in intelligent information processing from Xidian University, Xi'an, Shaanxi, China, in 2013.

He is a Professor with the School of Artificial Intelligence, Xidian University. He is also with Guangdong Artificial Intelligence and Digital Economy Laboratory, Guangzhou, Guangdong, China. His research interests include image processing, machine learning, and computational imaging.

Danhua Liu (Member, IEEE) received the Ph.D. degree in circuits and systems from Xidian University, Xi'an, Shaanxi, China, in 2010.

She is an Associate Professor with the School of Artificial Intelligence, Xidian University. Her research interests include compressive sensing, computational imaging, and machine learning.

Yanli Liu received the master's degree from China Academy of Space Technology in 2014. She is a Senior Engineer at Beijing Institute of Space Mechanics and Electricity. Her research focuses on the fields of computational spectral imaging and space optical remote sensing.

Guangming Shi (Fellow, IEEE) received the B.S. degree in automatic control, the M.S. degree in computer control, and the Ph.D. degree in electronic information technology from Xidian University, Xi'an, China, in 1985, 1988, and 2002, respectively.

He joined Xidian University in 1988. From 1994 to 1996, he was a Research Assistant with the Department of Electronic Engineering, The University of Hong Kong, Hong Kong. He has been a Professor with the School of Artificial Intelligence, Xidian University. His research interests include machine learning, compressed sensing, theory and design of multirate filter banks, image denoising, low-bit-rate image and video coding, and implementation of algorithms for intelligent signal processing.